

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)		Mass defect = $((235.01 + 1.01) - (91.90 + 140.89 + 3(1.01)))u$ [or $(235.01 - (91.90 + 140.89 + 2 \times 1.01)) = 0.20 u$ (1) Energy released = $0.20 \times 931 = 186 \text{ MeV}$ [190 MeV] (1) (Accept $3.0 \times 10^{-11} \text{ J}$) UNIT		2		2	2	
	(b)		General (G) G1: General shape of curve [correct asymmetry] & axes labelled G2: Maximum around nucleon number 60 / iron / nickel G3: Stability linked to BE/nuc to stability G4: Nucleons react to move towards maximum (on curve) G5: In doing so there is a loss / reduction in mass G6: Mass loss linked to energy released = $\Delta m c^2$ Fusion (Fu) Fu1: Smaller nucleon number nuclei combine Fu3: Larger nucleons of larger nucleon number formed: energy released, BE/nuc increases, stability increases, lower mass Fission (Fi) Fi1: Larger nucleon number split Fi3: Smaller nucleons of smaller nucleon number formed energy released, BE/nuc increases, stability increases, lower mass Additional point: gradient of curve larger in fusion region than fission resulting in more energy release (per nucleon) 5-6 marks At least 6 G points (6-8 G) 8 – 11 points <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i>	6			6		

		3-4 marks 5 – 7 points <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 – 4 points <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
		Question 4 total	6	2	0	8	2	0